Code[Coogs]

University of Houston

AI HACKATHON

Hacker Packet

Date: Friday, October 23, 2026

Time: 10:00 AM – 6:00 PM

Where: PGH – University of Houston

Format: Teams of up to 3 · Free to attend · Beginners welcome

This is your one-stop guide for the day - where to go, what happens when, both challenge tracks explained step by step, the rules, the prizes, and how to submit. New to hackathons or new to AI? This packet is written for you - no prior experience assumed.

Event Schedule

Here's the full day, start to finish. It's one track of time for everyone - you don't need to be anywhere else or sign up for separate sessions.

Time	What's Happening
10:00 AM	Check-in & Kickoff
10:30 AM	Hacking Begins
1:00 PM	Lunch
4:30 PM	Submissions Close
6:00 PM	Judging & Awards

What Each Block Means

- **Check-in & Kickoff (10:00 AM):** sign in, meet your team (or find one!), and hear a quick welcome and rundown of the two tracks.
- **Hacking Begins (10:30 AM):** this is when the clock starts - you and your team pick a track and start building.
- **Lunch (1:00 PM):** a break to eat, recharge, and keep working if you want - food is provided.
- **Submissions Close (4:30 PM):** your project needs to be submitted by this time - no exceptions, so plan to wrap up a little before.
- **Judging & Awards (6:00 PM):** judges review submissions and we announce the winners and hand out prizes.

Food & Drinks

Light snacks and drinks are available from check-in onward, lunch is provided at 1:00 PM, and snacks/drinks stay stocked throughout the day. Let us know about any allergies or dietary restrictions when you register, or flag an officer on the day - we'll do our best to accommodate.

Challenge Tracks

There are two challenge tracks this year. You'll pick one at check-in, and your team sticks with that track for judging.

The Same Basic Path, Either Track

Both tracks follow the same four steps: collect data, look for patterns, build and train a model, then use it to create something useful. The sections below walk through exactly what that means for each track.

Quick Definitions: “Build” vs. “Train”

You'll see “build a model” and “train a model” a lot in this packet - here's the difference, in plain terms:

- **Building** a model means choosing *how it works* - picking an approach (for example, a type of statistical method or machine learning algorithm) that fits the problem you're trying to solve.
- **Training** a model means *teaching it using your data* - showing it real examples so it can learn patterns and get better at making predictions.

In short: you build it once (decide how it works), then train it with data (teach it to actually work well). You'll do both on your track.

Track Overview

	Track 1	**Track 2**
Name	Hurricane Preparedness & Recovery	Houston Housing Market Intelligence
Category	Data Science & Predictive Analytics	Artificial Intelligence & Machine Learning
In one line	Help Houston prepare for the next storm.	Help Houstonians understand home prices.

Getting Started

Never built anything like this before? That's completely fine, this page is for you. Here are the tools people use, where to find data, and a simple way to get your first version working.

Tools We Recommend

- **Google Colab:** free, runs in your browser, and needs nothing installed on your laptop. The easiest way to start writing code and working with data.
- **Python:** the most common language for this kind of project, and one of the most beginner-friendly.
- **pandas:** a Python library for loading a dataset and exploring it as a table.
- **scikit-learn:** a beginner-friendly machine learning library with ready-made models, so you don't have to build one from scratch.
- **Notebooks (Colab or Jupyter):** let you run code in small pieces and see the result right away, which makes learning as you go much easier.

Where to Find Data

The dataset for your specific track will be shared at kickoff and posted in Discord, keep an eye out there. If you want to practice beforehand or explore on your own, these are good general sources:

- **Kaggle (kaggle.com):** free datasets on almost any topic, plus beginner tutorials.
- **Houston Open Data (data.houstontx.gov):** real data from the City of Houston.
- **Data.gov:** open datasets from the U.S. government.

A Simple Way to Start

Getting Started (continued)

1. Open a new Google Colab notebook (just go to colab.research.google.com and sign in).
2. Import pandas and load your dataset into it.
3. Use **.head()** and **.describe()** to look at what's actually in the data.
4. Ask simple questions about what you notice, you don't need a plan yet, just curiosity.
5. Pick one idea from your track's “Ideas to Get You Started” list and try the smallest version of it first.
6. Build from there. It's okay to start small and make it fancier later.

Stuck? Ask.

Officers will be walking around all day and are happy to help. Post questions in Discord any time. This is a learning event first, asking for help or pairing with your teammates is expected, not a sign you're behind.

Track 1: Hurricane Preparedness & Recovery

Data Science & Predictive Analytics

The Problem

Houston has experienced numerous hurricanes and extreme weather events that have caused widespread power outages, infrastructure damage, and financial losses. These events present major challenges for utility companies, emergency responders, and local governments when planning resources and recovery efforts.

Your Challenge

Using historical hurricane, weather, and other relevant datasets, develop a data-driven solution that helps Houston better predict, prepare for, respond to, or recover from future hurricanes. Explore patterns, build predictive models, and use your findings to propose a solution to a real-world problem.

Ideas to Get You Started

You can pick any angle - these are examples, not requirements:

- **Power Outage Prediction:** predict which areas are most vulnerable to outages and estimate restoration times.
- **Resource Optimization:** analyze how repair crews, equipment, and emergency resources could be distributed more efficiently.
- **Financial Impact:** estimate recovery costs for households, businesses, or city infrastructure.
- **Geographic Risk Analysis:** identify high-risk neighborhoods based on historical weather and geographic data.
- **Creative Solutions:** any other data-driven approach to hurricane preparedness and recovery.

What You're Expected to Show

You should demonstrate data science techniques - data analysis, statistical modeling, machine learning, or predictive analytics - to address your chosen problem. You have full freedom on methods, tools, and direction.

Your 4 Steps

Track 1: Hurricane Preparedness & Recovery (continued)

1. **Collect Data** - analyze past hurricanes, weather conditions, and power outages.
2. **Find Patterns** - identify relationships between storm severity, damage, and recovery times.
3. **Build & Train a Model** - choose an approach (build), then use statistics or machine learning to teach it with your data (train) so it can predict future impacts.
4. **Create Solutions** - help Houston improve emergency planning, resource deployment, and recovery.

Your goal, in one line: turn historical hurricane data into actionable insights that help Houston prepare for the next storm.

Track 2: Houston Housing Market Intelligence

Artificial Intelligence & Machine Learning

The Problem

Houston's housing market is constantly changing, with home prices varying based on location, size, and neighborhood. Understanding these differences can be challenging for homebuyers and residents. AI can help analyze housing data, predict home prices, and make housing information easier to understand.

Your Challenge

Using a Houston housing dataset (exact dataset to be confirmed), train an AI model to predict home prices based on property information. You're encouraged to go beyond price prediction and create an innovative solution that helps Houston residents better understand housing costs and make informed decisions.

Ideas to Get You Started

You can pick any angle - these are examples, not requirements:

- **Housing Price Prediction:** use AI to estimate home prices.
- **Affordable Housing:** help residents find homes within their budgets.
- **Property Value Analysis:** identify potentially overpriced or underpriced homes.
- **Neighborhood Comparisons:** compare housing prices across Houston neighborhoods.
- **Creative Solutions:** any other AI-powered tool that addresses housing challenges.

What You're Expected to Show

You must train and improve an AI or machine learning model using housing data. You have freedom on tools and approach, but model training needs to be a central part of your project.

Your 4 Steps

Track 2: Houston Housing Market Intelligence (continued)

1. **Collect Data** - pull in a Houston housing dataset (property info, location, size, price).
2. **Explore Patterns** - look at how price relates to location, size, and neighborhood.
3. **Build & Train a Model** - choose an approach (build), then teach an AI/ML model with the housing data (train) so it can predict home prices.
4. **Build Your Solution** - turn predictions into something useful, like an affordability tool, a price-comparison view, or an over/underpriced-home flagger.

Your goal, in one line: use AI to help Houstonians understand home prices.

Housing affordability affects a lot of Houston residents, especially families, students, and first-time homebuyers - this track is a chance to make housing information more accessible and support better decisions.

Prizes & Rules

Prizes

- **1st Place:** MSI Monitor + medal
- **2nd Place:** Mechanical Keyboard + medal

Medals go to the teams that place.

Ground Rules

1. Hack anywhere you want - just be back in time for submission.
2. Teams of up to 3 - solo, duo, or trio are all welcome.
3. Pick one track and stick with it for judging.
4. No late entries - submit before the deadline.
5. Keep it respectful and be mindful of everyone around you.

Judging, Submission & Contacts

How Judging Works

Judges will be looking at a few simple things - the exact scoring rubric will be shared by the judging team on the day, but here's what matters:

- **Does it work?** How far did you get, and how well does it run?
- **Is it creative?** Is the idea or approach fresh?
- **Is it useful?** Would this matter to real people outside the hackathon?
- **Can you explain it?** Can your team walk judges through what you built and why, clearly?

How to Submit

No file or link to upload, when submissions close, you'll present straight to the judges. Just have these two things ready:

- **A demo** of what you built, ready to show live.
- **A pitch** explaining what you built and why it matters, a couple minutes is plenty.

Be ready by 4:30 PM, no late entries.

Join the Discord

This is where all day-of updates, announcements, and support happen - join before the event if you can: **discord.gg/74CagA9Rt**

Officer Contacts

Jonathan Cummins	President
Andy Dao	Competition Director
Jeric Panugaling	Competition Officer
Dr. Nouhad Rizk	Founder & Faculty Advisor

General questions: codecoogs@gmail.com · codecoogs.com · @uh_codecoogs